

Answer Key

Question 1

a) **Sample Space:**

For a fair coin toss, $\Omega = \{H, T\}$.

b) **Disjoint Events:**

Events A (“Head”) and B (“Tail”) are disjoint because they cannot occur simultaneously in a single coin toss.

Question 2

We know:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

So,

a)

$$P(A \cap B) = 0.4 + 0.5 - 0.7 = 0.2.$$

b)

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.7 = 0.3.$$

Question 3

a) **Sample Space:**

$$\{HH, HT, TH, TT\}.$$

b) **Event C :** “Exactly one head” corresponds to $\{HT, TH\}$.

Thus,

$$P(C) = \frac{2}{4} = 0.5.$$

c) **Independence Check:**

- Let F = “first toss is a head” with outcomes $\{HH, HT\}$, so $P(F) = \frac{2}{4} = 0.5$.
- Intersection $F \cap C = \{HT\}$ with $P(F \cap C) = \frac{1}{4} = 0.25$.

- Since $P(F)P(C) = 0.5 \times 0.5 = 0.25$ and $P(F \cap C) = 0.25$, the events are independent.
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Question 4

$X \sim \text{Binomial}(10, 0.3)$.

a)

$$P(X = 3) = \binom{10}{3} (0.3)^3 (0.7)^7.$$

Calculation:

$$\binom{10}{3} = 120,$$

$$(0.3)^3 = 0.027,$$

$$(0.7)^7 \approx 0.08235.$$

Thus,

$$P(X = 3) \approx 120 \times 0.027 \times 0.08235 \approx 0.2668.$$

b)

$$E[X] = np = 10 \times 0.3 = 3,$$

$$\text{Var}(X) = np(1 - p) = 10 \times 0.3 \times 0.7 = 2.1.$$

Question 5

$Y \sim \text{Uniform}(2, 8)$.

a) **PDF:**

$$f_Y(y) = \frac{1}{8 - 2} = \frac{1}{6}, \quad 2 \leq y \leq 8.$$

b)

$$P(3 \leq Y \leq 5) = \int_3^5 \frac{1}{6} dy = \frac{5 - 3}{6} = \frac{2}{6} = \frac{1}{3} \approx 0.3333.$$

Question 6

$X \sim N(100, 16^2)$.

a) **Z-scores:**

For $X = 90$:

$$z = \frac{90 - 100}{16} = -\frac{10}{16} = -0.625.$$

For $X = 110$:

$$z = \frac{110 - 100}{16} = \frac{10}{16} = 0.625.$$

b)

$$P(90 \leq X \leq 110) = P(-0.625 \leq Z \leq 0.625).$$

Using standard normal tables:

- $P(Z \leq 0.625) \approx 0.7340$,
- $P(Z \leq -0.625) \approx 0.2660$.

Thus,

$$P(90 \leq X \leq 110) \approx 0.7340 - 0.2660 = 0.4680.$$

Question 7

Population parameters: $\mu = 10$ and $\sigma^2 = 9$.

a) **Sampling Distribution:**

For a sample of size $n = 36$, the sampling distribution of $\bar{X}$ is:

$$\bar{X} \sim N\left(10, \frac{9}{36}\right) = N(10, 0.25).$$

b) Standard error: $\sigma_{\bar{X}} = \sqrt{0.25} = 0.5$.

Compute:

$$P(\bar{X} \leq 11) = P\left(\frac{\bar{X} - 10}{0.5} \leq \frac{11 - 10}{0.5}\right) = P(Z \leq 2).$$

From the Z-table, $P(Z \leq 2) \approx 0.9772$.

Question 8

For $n = 25$, $\bar{X} = 50$, $s = 4$, degrees of freedom = 24.

Using the t-distribution, the 95% confidence interval is:

$$\bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}.$$

From t-tables, $t_{0.025,24} \approx 2.064$.

Margin of error:

$$2.064 \times \frac{4}{5} = 2.064 \times 0.8 \approx 1.6512.$$

Thus, the 95% CI for μ is approximately:

$$(50 - 1.65, 50 + 1.65) = (48.35, 51.65).$$

Question 9

A **parameter** is a numerical characteristic of a population (e.g., the true average income of all residents in a city, say $\mu = 60,000$ dollars). A **statistic** is a numerical measure computed from a sample (e.g., the sample mean income calculated from 200 randomly selected residents). The statistic is used to estimate the parameter.

Question 10

a) **Distribution Identification:**

The number of products meeting the standard follows a Binomial distribution:

$$X \sim \text{Binomial}(n = 100, p = 0.7).$$

b)

$$P(X = 75) = \binom{100}{75} (0.7)^{75} (0.3)^{25}.$$

(Leave the answer in this exact form or compute numerically if required.)

c) **Normal Approximation:**

For $X \sim \text{Binomial}(100, 0.7)$:

- Mean: $\mu = np = 70$,
- Variance: $\sigma^2 = np(1 - p) = 21$,
- Standard deviation: $\sigma \approx \sqrt{21} \approx 4.583$.

For $P(X \geq 68)$, use continuity correction:

$$P(X \geq 68) \approx P(Y \geq 67.5),$$

where Y is $N(70, 4.583^2)$.

Compute the z-score:

$$z = \frac{67.5 - 70}{4.583} \approx \frac{-2.5}{4.583} \approx -0.546.$$

Then,

$$P(Y \geq 67.5) = 1 - P(Z \leq -0.546) = 1 - 0.292 \approx 0.708.$$

Thus, the approximate probability is about 70.8%.